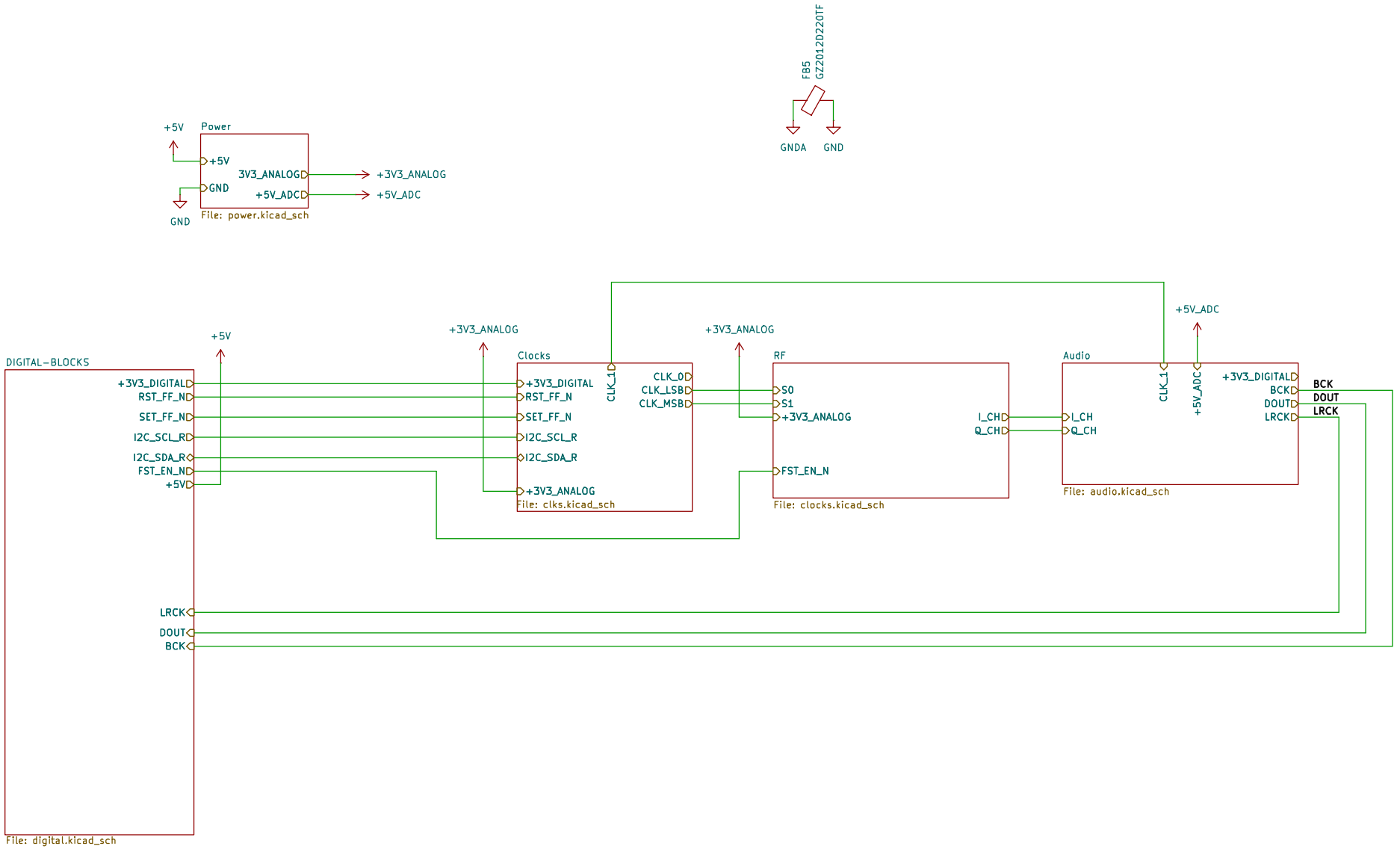
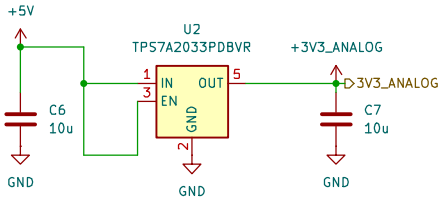
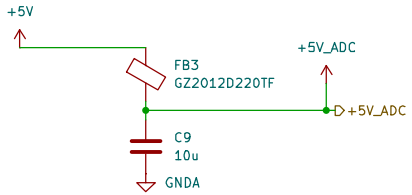
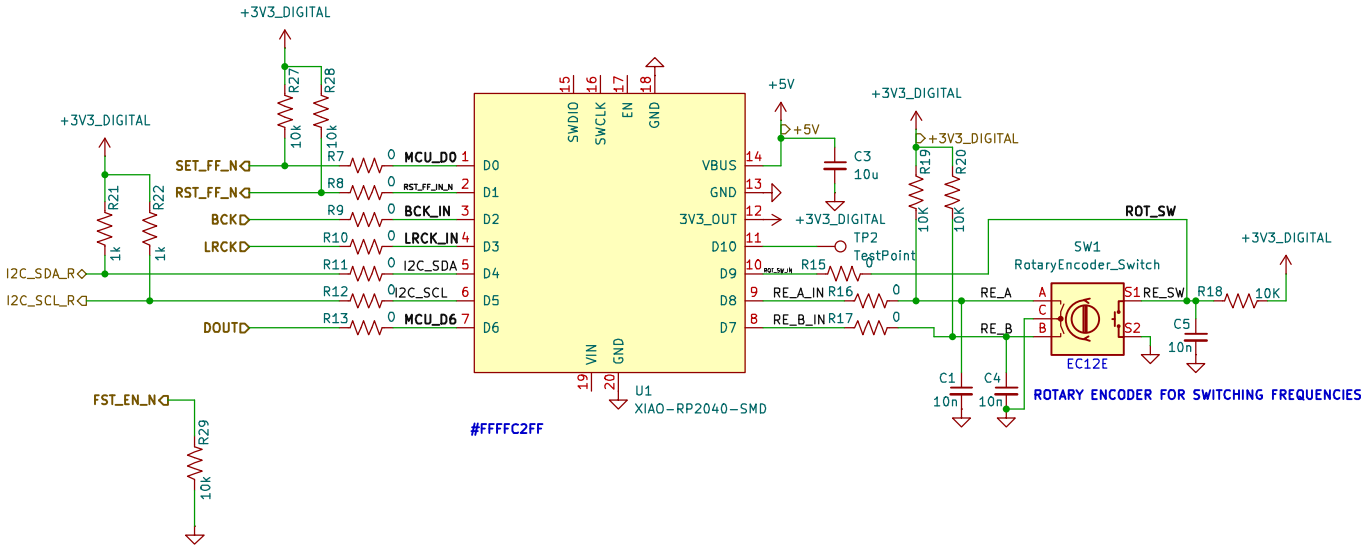
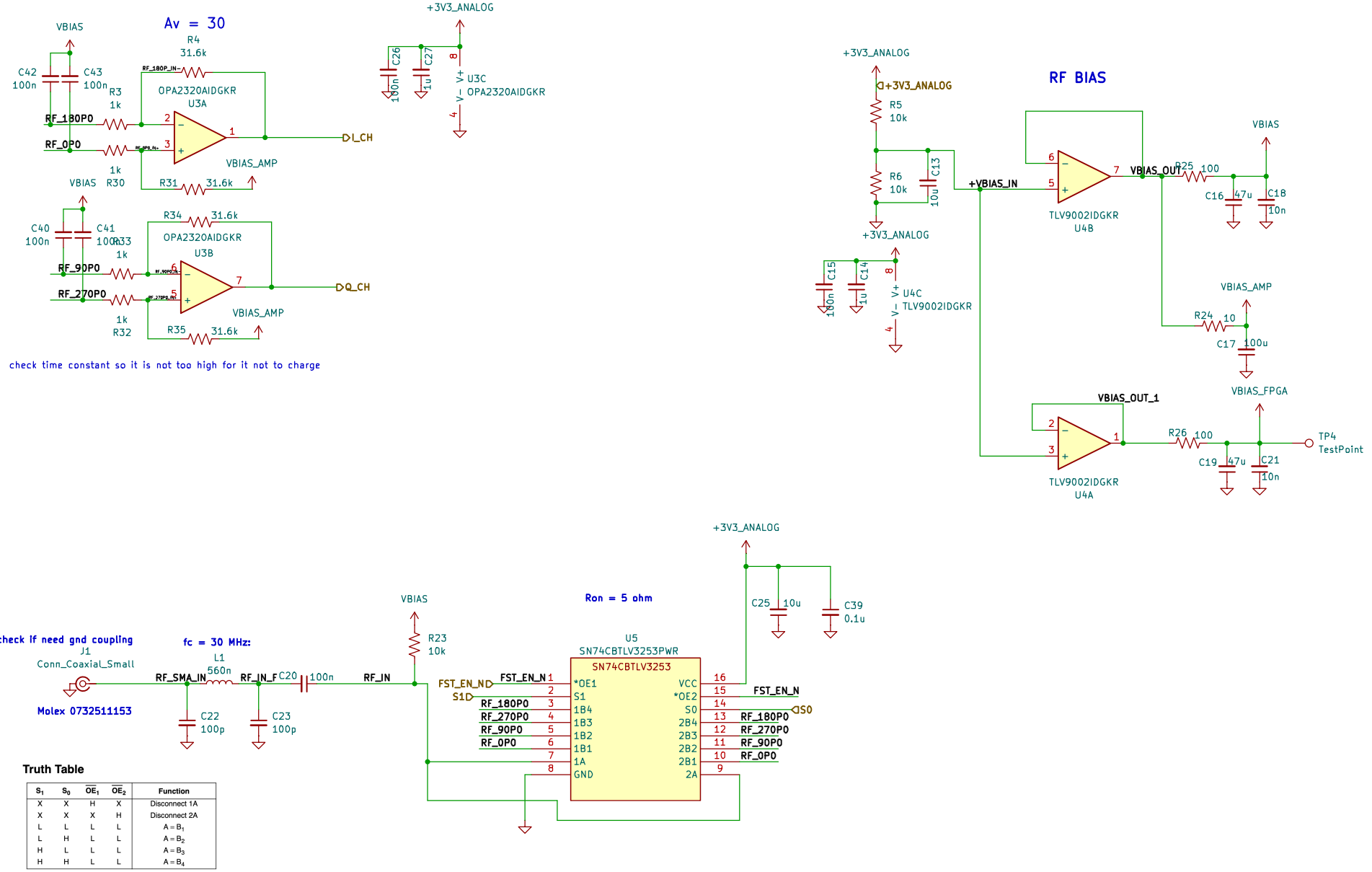

FB5
GZ2012D220TF
GNDA
GND
+5V
Power
+5V
3V3_ANALOG
GND
+5V_ADC
+3V3_ANALOG
+5V_ADC
GND
File: power.kicad_sch
+5V_ADC
+5V
+3V3_ANALOG
+3V3_ANALOG
DIGITAL-BLOCKS
Clocks
RF
Audio
+3V3_DIGITAL
RST_FF_N
SET_FF_N
I2C_SCL_R
I2C_SDA_R
FST_EN_N
+5V
+3V3_DIGITAL
RST_FF_N
SET_FF_N
I2C_SCL_R
I2C_SDA_R
+3V3_ANALOG
CLK_1
CLK_0
CLK_LSB
CLK_MSB
S0
S1
+3V3_ANALOG
FST_EN_N
I_CH
Q_CH
I_CH
Q_CH
CLK_1
+5V_ADC
+3V3_DIGITAL
BCK
DOUT
LRCK
BCK
DOUT
LRCK
LRCK
DOUT
BCK
File: clks.kicad_sch
File: clocks.kicad_sch
File: audio.kicad_sch
File: digital.kicad_sch


+5V
U2
TPS7A2033PDBVR
+3V3_ANALOG
1
IN
OUT
5
3
EN
GND
2
3V3_ANALOG
C6
10u
C7
10u
GND
GND
GND


+5V
+5V_ADC
FB3
GZ2012D220TF
+5V_ADC
C9
10u
GNDA


+3V3_DIGITAL
R27
10k
R28
10k
+3V3_DIGITAL
R21
1k
R22
1k
SET_FF_N
RST_FF_N
BCKD
LRCKD
I2C_SDA_R
I2C_SCL_R
DOUTD
R7
R8
R9
R10
R11
R12
R13
0
MCU_D0
RST_FF_IN_N
BCK_IN
LRCK_IN
I2C_SDA
I2C_SCL
MCU_D6
SWDIO
SWCLK
EN
GND
15
16
17
18
D0
D1
D2
D3
D4
D5
D6
1
2
3
4
5
6
7
VBUS
GND
3V3_OUT
D10
D9
D8
D7
14
13
12
11
10
9
8
+5V
C3
10u
+3V3_DIGITAL
TP2
TestPoint
R15
R16
R17
RE_A_IN
RE_B_IN
VIN
GND
19
20
U1
XIAO-RP2040-SMD
#FFFFC2FF
+5V
+3V3_DIGITAL
R19
10K
R20
10K
ROT_SW
SW1
RotaryEncoder_Switch
RE_A
RE_B
A
C
B
S1
S2
EC12E
RE_SW
R18
10K
+3V3_DIGITAL
C5
10n
C1
10n
C4
10n
ROTARY ENCODER FOR SWITCHING FREQUENCIES
FST_EN_N
R29
10k


Av = 30
VBIAS
R4
31.6k
C42
100n
C43
100n
R3
1k
RF_180P0
RF_0P0
OPA2320AIDGKR
U3A
VBIAS_AMP
I_CH
R30
1k
R31
31.6k
+3V3_ANALOG
C26
100n
C27
1u
U3C
OPA2320AIDGKR
C40
100n
C41
100n
R33
1k
R34
31.6k
RF_90P0
RF_270P0
OPA2320AIDGKR
U3B
Q_CH
R32
1k
R35
31.6k
check time constant so it is not too high for it not to charge
RF BIAS
R5
10k
R6
10k
C13
10u
+VBIAS_IN
TLV9002IDGKR
U4B
VBIAS_OUT
R25
100
C16
47u
C18
10n
C15
100n
C14
1u
U4C
TLV9002IDGKR
VBIAS_AMP
R24
10
C17
100u
VBIAS_FPGA
VBIAS_OUT_1
R26
100
C19
47u
C21
10n
TP4
TestPoint
TLV9002IDGKR
U4A
check if need gnd coupling
J1
Conn_Coaxial_Small
Molex 0732511153
fc = 30 MHz:
L1
560n
RF_SMA_IN
RF_IN_F
C20
100n
RF_IN
C22
100p
C23
100p
R23
10k
Ron = 5 ohm
U5
SN74CBTLV3253PWR
SN74CBTLV3253
C25
10u
C39
0.1u
FST_EN_N
S1D
S0
RF_180P0
RF_270P0
RF_90P0
RF_0P0
Truth Table
S1 S0 OE1 OE2 Function
X X H X Disconnect 1A
X X X H Disconnect 2A
L L L L A = B1
L H L L A = B2
H L L L A = B3
H H L L A = B4

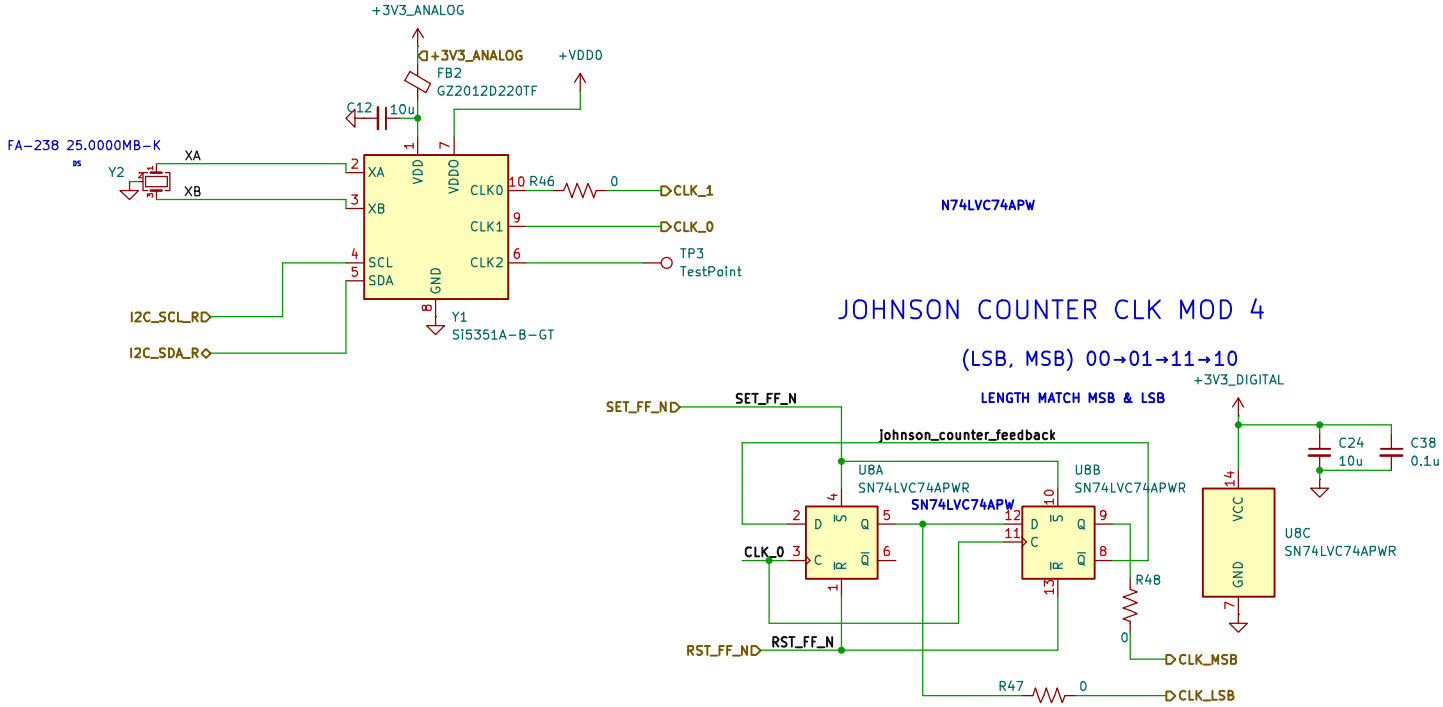

+3V3_ANALOG
+3V3_ANALOG
FB2
GZ2012D220TF
+VDDO
C12
10u
FA-238 25.0000MB-K
XA
Y2
XB
XA
XB
SCL
SDA
VDD
VDDO
GND
CLK0
CLK1
CLK2
R46
0
CLK_1
CLK_0
TP3
TestPoint
I2C_SCL_RD
I2C_SDA_R
Y1
SI5351A-B-GT
N74LVC74APW
JOHNSON COUNTER CLK MOD 4
(LSB, MSB) 00→01→11→10
LENGTH MATCH MSB & LSB
SET_FF_ND
SET_FF_N
johnson_counter_feedback
U8A
SN74LVC74APWR
SN74LVC74APW
U8B
SN74LVC74APWR
U8C
SN74LVC74APWR
+3V3_DIGITAL
C24
10u
C38
0.1u
CLK_0
R48
0
CLK_MSB
RST_FF_ND
RST_FF_N
R47
0
CLK_LSB

Table 2. Interface Modes

MD1 (PIN 11)	MD0 (PIN 10)	INTERFACE MODE
Low	Low	Slave mode (256 f_S, 384 f_S, 512 f_S autodetection)
Low	High	Master mode (512 f_S)
High	Low	Master mode (384 f_S)
High	High	Master mode (256 f_S)

Table 3. Data Format

FORMAT NO.	FMT (Pin 12)	FORMAT
0	Low	I^2S, 24-bit
1	High	Left-justified, 24-bit

